

Learning SPS Through Its Literature Lineage

A worked technical reading

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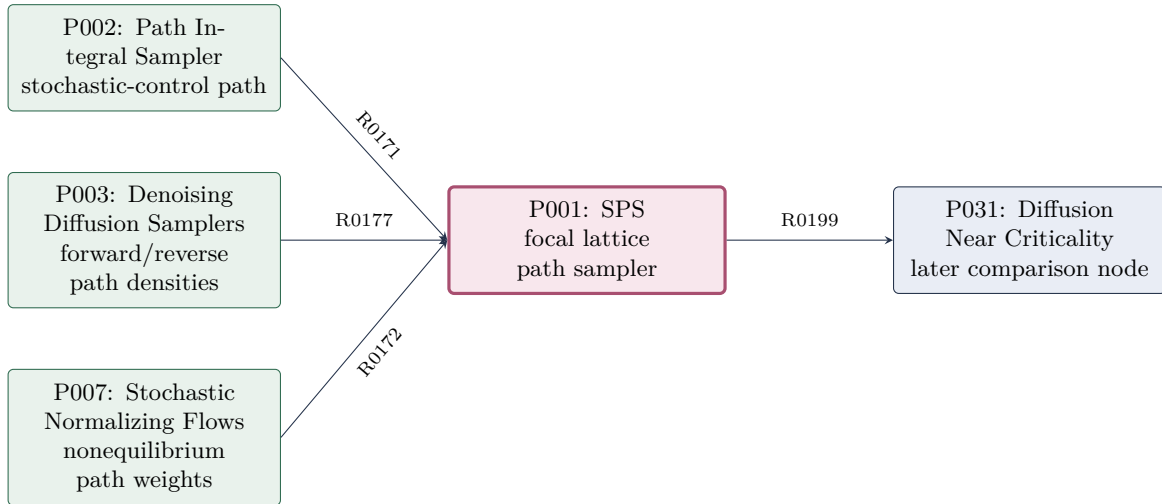
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Draft status. The SPS paper is mapped at equation level; comparisons with P002, P003, and P007 are still in progress.

1 Learning Question

Which earlier methods are needed to understand SPS, how does SPS work, and which technical differences are supported by the current sources? The target is T3: reconstruct the ordered algorithm after explaining its core formulas. Code mapping and reproduction are not requested.

2 Lineage-First Route



Edge meaning. R0171, R0172, R0177, and R0199 are checked direct-citation relations. Dijkstra costs set the reading order; the technical claims use the cited source sections.

3 Finding The Technical Difference

Step	Read first	Purpose
I0	exact P001 version and source hash	lock the object
I1	contribution paragraph and overview	record the authors' claimed delta
I2	matching P002/P003/P007 equations	subtract inherited components
I3	P001 Eqs. (2.2)–(2.19)	isolate the operative mechanism change
I4	P001 Tables 1–4 and Fig. 8	ask which evidence isolates the change
I5	P031 method and direct-citation context	test persistence without inferring adoption

The current candidates are the paired forward/backward Langevin parameterization, the path-space log-ratio objective, and full-trajectory IMH. These mechanisms appear in P001, but their novelty rank remains **pending** until the predecessor equations are aligned.

4 Technical Mechanism

1. Sample the initial field from a tractable prior.
2. Generate a forward Langevin path with a learned drift (P001 p. 5, Eq. (2.2), E-260613790-M).
3. Evaluate an auxiliary backward path on the same trajectory (P001 pp. 5–6, Eqs. (2.3), (2.7)–(2.8)). Evidence: E-260613790-M.
4. Minimize the expected forward/backward path log-ratio (P001 p. 6, Eq. (2.13), E-260613790-M).
5. Propose independent full trajectories and apply trajectory-level IMH (P001 pp. 7–9, Eqs. (2.18)–(2.19), E-260613790-M).
6. Estimate observables and autocorrelation from the corrected chain.

5 Core Formulas And Limits

$$s_{i+1} = s_i + \sigma_\theta^2(t_i) K_{\theta,F}(s_i, t_i) \Delta t + \sigma_\theta(t_i) \xi_i \sqrt{\Delta t}, \quad (\text{F01})$$

$$D_{\text{KL}} = \int dq_F(\tau) \log \frac{dq_F(\tau)}{d\tilde{q}_B(\tau)}, \quad (\text{F02})$$

$$P_{\text{acc}} = \min \left(1, \frac{\tilde{\pi}^*(s'_T) T(s'_T \rightarrow s_T)}{\tilde{\pi}^*(s_T) T(s_T \rightarrow s'_T)} \right). \quad (\text{F03})$$

F01 generates the path, F02 trains it without the target normalization constant, and F03 corrects a support-covering proposal. Finite model capacity, training, and discretization leave residual irreversibility before correction (P001 pp. 7–8, E-260613790-L).

6 Evidence And Technical Review

Claim	What the source supports	Limit
Corrected observables	Agreement with the reported HMC values in the tested 2D $L \times 8$ scan (P001 Figs. 3–5, Tables 1–3, E-260613790-R)	benchmark-specific
Autocorrelation	About 0.5 for SPS+IMH and about 160 for HMC in the reported $L = 64$, $\kappa = 0.27$ comparison (P001 Fig. 8, E-260613790-R)	different update units; not a cost ratio
Timing	About 1.1–1.2 h training and 1.1–2.1 min generation across the reported sizes (P001 Table 4, E-260613790-L)	no matched HMC GPU-hour comparison
Exactness	trajectory-level IMH restores target invariance (E-260613790-M)	requires target-support coverage

7 Learning Status

Level	Evidence	Status
T0	exact version, URL, hash, and identity rows	pass
T1	explanation preserving result and cost limits	pass
T2	F01–F04 with symbols, roles, assumptions, and anchors	pass
T3	focal algorithm plus predecessor comparison	pending
T4	official equation-to-code mapping	not requested
T5	minimal reproduction	not requested

Next action. Upgrade P002, P003, and P007 to detailed R0–R6 records; align their path objectives and correction rules with P001; then update the innovation table and technical review. The current conclusion covers the paired-path lattice mechanism and its trajectory-level correction; novelty and cost comparisons remain open.